

0107E801-3

Candidate's Seat No: _____

**IM.Sc. (CS) Sem.-8 Examination
Machine Learning**

Time : 3.00 Hours]

July-2025

[Max.Marks : 70

Instructions:

- **Write both the Sections in the separate answer book.**
- **Both Sections having equal weightage.**
- **Draw Diagrams wherever necessary.**
- **Make Assumptions wherever necessary.**

SECTION – I

Q.1 Answer the following questions: 4

[15]

- (a) Explain the concepts of bias and variance in machine learning. How do they affect model performance?
- (b) Compare the Elbow Method and Silhouette Score for determining the optimal number of clusters in K-means.
- (c) Convert the word "INTENTION" to "EXECUTION" using the minimum number of edit operations.

OR

- (c) Compute the cosine similarity for the following documents:

Document 1: "machine learning algorithms are easy"

Document 2: "Algorithms for learning machines are hard"

Q.2 Attempt the following (Any TWO) 10

[20]

- (a) What are the different types of data used in machine learning? Explain with examples.
- (b) Difference Simple Linear Regression and Multiple Linear Regression? Derived the equation for slope and intercept of best fit line in Simple Linear Regression.
- (c) What is hierarchical clustering? Explain the difference between agglomerative and divisive clustering. Also, describe the different linkage methods used in hierarchical clustering.

(P.T.O)

SECTION – II

Q.3 Answer the following questions: 5

[15]

- (a) What is the purpose of the Attribute Selection Measure (ASM) in decision trees? Describe various Attribute Selection Measure method.
- (b) What is the kernel trick, and why is it important in SVM?

OR

- (b) Compare the Linear Kernel, Polynomial Kernel, and RBF Kernel in terms of their use cases.
- (c) Define the following terms in SVM:
- Hyperplane
 - Support Vectors
 - Margin
 - Kernel
 - Regularization (C)

Q.4 Attempt the following (Any TWO)°

[20]

- (a) What is Ensemble Learning. Explain various techniques in Ensemble Learning.
- (b) What is a Random Forest algorithm? Explain its working with steps and advantages. Why is it considered more accurate than a single decision tree?
- (c) Given the following linearly separable 2D dataset:

Points	Class
(1, 2)	Circle
(2, 3)	Circle
(3, 1)	Circle
(0, 0)	Square
(1, 1)	Square
(2, 0)	Square

Let Support vectors are identified as (1, 2), (2, 0), and (1, 1) then compute:

- a) The weight vector w
- b) The bias term b
- c) The equation of the optimal hyperplane
- d) Identify the label of point (2,2).